"LISTENING AND REMEMBERING": NETWORKED OFF-LINE IMPROVISATION FOR FOUR COMMUTERS

Ximena Alarcón
Institute of Creative Technologies, IOCT, De Montfort University
XAlarcon@dmu.ac.uk
The Gateway, Leicester LE1 9BH, UK

ABSTRACT

This paper analyses the experience of the networked off-line improvisation "Listening and Remembering", a performance for four commuters using voices and sounds from the México DF and Paris metros. It addresses the question: how can an act of collective remembering [20], inspired by listening to metro soundscapes, lead to the creation of networked voice- and sound-based narratives about the urban commuting experience?. This hybrid sonic performance extends characteristics of the Interconnected Music Network art-form [16] by adding the emotional preparation of participants (through an ethnographic process) and spatialisation. When commuters follow the structure of a journey in the metro through this networked environment, the result is a soundscape involving a counterpoint of voices, recorded voices, and machinery, all together eliciting diverse narratives. Participants re-enact their routine experience through a dialogical relationship with the sounds, themselves and the experience of sharing: a collective memory.

1. BACKGROUND

In 2003, I undertook ethnographic research about memories of soundscape with commuters of the London Underground [1], while reflecting on the intersection between the theory and concepts of soundscape [14] and of collective memory [20]. Through interviews, recordings of journeys, and listening[1] practices, I investigated commuters' feelings triggered by sound within their traveling routine. It led to the creation of an Internet-based *Interactive Sonic Environment* [2], where commuters interact with the soundscape, mediated by a process of remembering[2]. Users experience a non-linear journey in the underground by interacting with commuter-selected sounds of commuters' journeys, which overlap in a multimedia score composed of spaces, categories and subspaces. I proposed "Listening and Remembering" as the main interactive practice and it was initially fulfilled by on-line text responses to the listening experience [1,2].

Within the sonic environment and during the different research stages participants identified 'voice' as one of the most attractive features of this contemporary underground soundscape. They often miss it as a dialogical and expressive instrument during their commuting routine. Thus, voice became an important element to re-enact the human experience in this environment, and it provided a opportunity, in this context, for making Wertsch's idea of "voices of collective remembering"[3] *alive* and *heard*. Wertsch has called "textual community", to a group using a set of cultural tools, language, and objects to produce narratives or "a specific type of community, - namely, one grounded in the use of a shared set of texts."[22] This concept of textual community could be linked with Truax's concept of acoustic community. However, these concepts differ in the sense that Wertsch's community is created by sharing texts and producing narratives while Truax's community shares the acoustic environment, no matter how "this commonality is understood"[15]. Wertsch leaves "voice" as translated usually as "text" in the practice of remembering. Truax, leaves the existence of a community without specific artistic practice for exchanging sounds or feelings attached to them, thus to create community.

My current research project, called *"Sounding Underground"*[4], continues the exploration of sound stories of the commuting experience, with a particular interest in creating a space for commuters' expression and interweaving of memories with their voice. It investigates the links among the soundscapes of the México DF Metro, the Paris Metro, and the London Underground, while emphasizing the uniqueness of each underground transport system. The research consolidates my previous ethnographic methodology of interviewing, recording, listening, and selecting

[1] Listening is understood here in its wider sense as "a complex, multi-layered activity of which hearing is but a part…References, memories, associations, symbols, - all contribute to our understanding of sonic meaning." [8]

[2] According to James Werstch the process of remembering is related to the "textual resources" utilised in talking or writing about the past, to how those texts could show an image of the view of others who have utilised them, and, eventually, to the addition of the voice of whoever is recalling. [21]

[3] "Voices of Collective Remembering" is the title of James Werstch's book where concept of collective memory, textual communities and the process of remembering, are explained. [20]

[4] The project originally was called "Linking soundscapes via commuters' memories". It has evolved during the course of the project, and the new working title is "Sounding Underground".

sounds, and introduces improvisation as a performance activity.

2. CONTEXT

The soundscape of London Underground (LU) is a rich mixture between sonic, symbolic, and social contexts. During the journey, repetitive sounds of machinery, reverberant spaces, contrasts between confined and open spaces, and the everyday interaction of people with a purposeful activity, have created sophisticated sound textures and rhythms, full of micro-events, from which to approach both unconscious and conscious states of commuters' bodies and minds [1]. During the journey, a powerful connection with the individual's own life is provoked by this magical and sublime space. Williams refers to the aesthetic concepts of sublimity and fantasy as "invented to express the emotional power of subterranean environments, a power not encompassed by the traditional aesthetic terminology of beauty and ugliness"[24]. Sublimity in underground environments "celebrates ambivalence" depending on "the delicate equipoise of conflicting emotions" [25]. This experience has inspired humans and their fantasy since the first attempts of exploitation and industrialization of underground environments [23]. Symbolically the underground journey can represent death [6, 26], the disconnection from our known environment and natural light; also the journey can be thought of as re-birth, the connection to ourselves in a space of detachment, in a "womb-like" space. In the *isolation* of a public space, guided by the rhythm and movement of machines and people, conditions are created for *digging* into profound human emotions. Comparing the LU's commuters' experience with the one of commuters in México DF and Paris, it is noticeable that socially, each city's underground unveils tensions, as regards class, gender, social status, political control, identity and belonging to a particular community. These relationships are immersed in the cultural appropriation of a technological environment [23].

In the ethnographic process, seventeen volunteers in México DF and seventeen in Paris participated in an individual process of remembering that started with an interview about their commuting experience, and in particular about the remembered sounds and feelings associated with it. Secondly, each participant recorded a journey in the metro, using binaural microphones, experiencing an awareness of details of the soundscape not previously perceived, as well as of their own role within the soundscape.

During the recording, the researcher offered the participants small papers containing messages. These invited them to interact, think, or perceive in particular ways, sonic and memorable aspects of their journey, creating a playful intervention. Some of the participants became engaged with this interaction, commenting, or reflecting in silence, intrigued by the chance situations that occurred when the message was read. Others continued with their routine, as was also their option, while recording their journey.

In a subsequent session, participants listened to their journeys and selected and edited[5] the sounds that they considered most meaningful. The only constraints imposed by the project were the duration of the sounds: 15" at the shortest, and 1'30" at the longest.

The improvisation was the last stage in this series of workshops during the fieldwork. It functioned as a compilation of the entire experience of remembering. The previous fieldwork activities were considered essential emotional preparation for the improvisation.

3. NETWORKED IMPROVISATION

Networked music performances are mostly performed by trained musicians and performers, as such performances traditionally demand musical and technical skills [19]. Barbosa, in his classification of "computer-supported collaborative music", points out that "a spontaneous [free] improvisational approach" [3] is suitable in an Internet context, because of the characteristics of this medium, and introduced the term *Shared Sonic Environments* to describe a kind of performance where people can "participate in a public event by manipulating or transforming sounds and musical structures or by simply listening to music created collectively."[3] This "public event" implies the participation of non-musicians and suggests the creation of acoustic communities via the Internet, going "beyond the enhancements of existing acoustic communication paradigms"[4]. On the other hand, listening in an electroacoustic musical space to environmental sounds implies the expansion of the listening experience, "transporting the listener beyond the listening space or creating a large space for the listener to inhabit"[12].

Thus, the "Listening and Remembering" improvisation is described here as a *Shared Sonic Environment* that extends the perception of space by the use of spatialisation of soundscape sounds and the voices of commuters expressing memories. At the same time, it uses an on-line web 2.0 model of sharing, introducing tags as a resource to identify, track and trigger participants' recordings. Most of the participants had no performance experience. Their knowledge of the soundscape and what this means for their lives were the most important elements brought into the performance space. In terms of geographical location, the improvisation is off-line, meaning a co-located system[6] where participants' computers are linked by a local area network, and share the same

[5] I gave a tutorial in Audacity that taught how to excerpt sounds and make fades.

[6] A co-located system is defined by Barbosa [3] as a networked performance in which participants share the same physical space. He derives this definition from Rodden's Computer Support for Cooperative Working (CSCW) geographical nature dimension [10]. The term helps to define the "off-line" character of this networked improvisation, meaning also that it is not connected to the Internet.

room. It has been implemented first off-line (co-located), in order to establish a model for what could be created on-line (remote), thus including commuters from the other metros.

In order to describe different aspects of the improvisation, I employ Weinberg's theoretical framework for Interconnected Musical Networks [16]. The four main aspects that she describes—goals and motivation, musical content and control, social organization and perspectives, and architectures and topologies—are useful for understanding this performance experience as "an interdependent art form", while they simultaneously present a foil for the ways in which it represents a new approach to networked musical experiences.

In terms of 'goals and motivation', it can be understood as an *exploratory* and *process*-centered network. For Weinberg, exploratory networks "do not impose specific directions or goals for the participants" [17]. This improvisation maintains the narrative structure of an underground journey. I proposed to groups of four participants, whose computers were networked, that they listen to a journey based on sonic excerpts of their own journeys, organized and triggered according to such a narrative, by means of *environments* (i.e. Street, Entrance, Tickets, Corridors, Platform, Carriage), and sonic intervals called *events* (i.e. steps, amplified voices, trains arriving, opening and closing doors). The sounds were diffused without any transformation via five loudspeakers that surrounded the participants (see Figures 1 and 2): participants were each allocated an individual speaker through which their memories would be heard, while environments and events were deployed to multiple speakers, presenting a wider acoustic space. When they wanted to express a memory[7], they were able to record it via the microphone, and each memory was given a name, or tagged (see Figure 3), to be visible and available for playback on the other three participants' screens. The responses of the participants were unpredictable and the space was free for them to express as they wished.

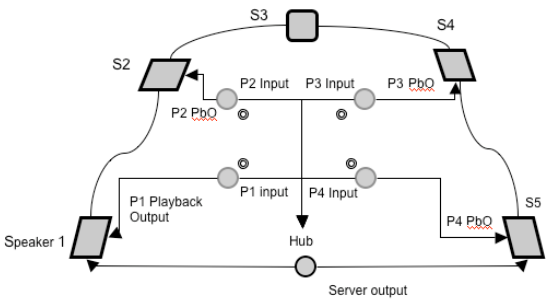


Figure 1. Topology.

[7] Memory, within this context, is understood as any form of expression made with the voice, with or without words, that has been triggered by the soundscape.



Figure 2. Participants improvising in Paris

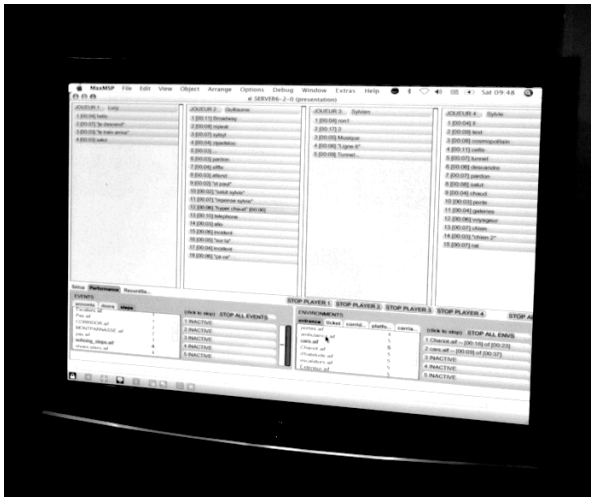


Figure 3. Server patch in improvisation in Paris.

Thus, the 'musical content' we hear during the improvisation is a mixture of commuters' journeys, including counterpoint between human and machinery sounds, and the responses that these provoke. I, through the server software, control the amplitude and triggering of pre-recorded sound excerpts. The commuters, through the client software, control the recording and playback of the voices. As there is no transformation of sounds, all sonic events and their combination depend on human input. All these aspects define two actors in terms of 'social organization and perspectives': the leader (i.e. the researcher) and the participants (i.e. the commuters).

The technical setting, created by the composer and Max/MSP programmer Peter Batchelor, consisted of five computers connected via Ethernet. Two Max/MSP patches were designed: one for the leader of the improvisation - the server - (see Figures 4, and 5), and one for the participants - the clients - (see Figure 6). Their interfaces were designed to be colourful and engaging while being as intuitive and easy to use as possible, in order to enable fluency within the improvisation. The server acted as a hub for all audio activity, receiving controller data (record/playback triggers) from the clients.

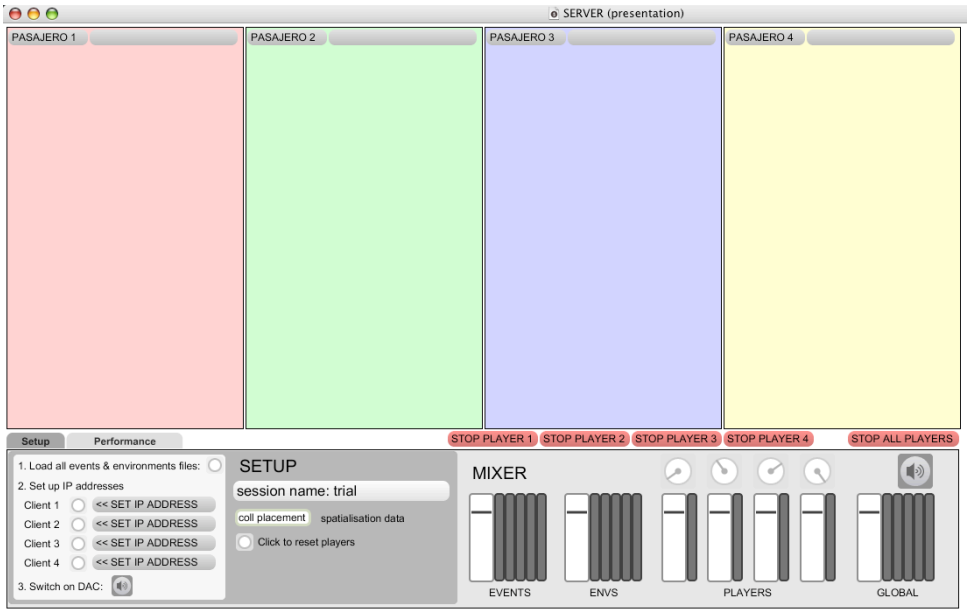


Figure 4. Server patch with features of the Setup.



Figure 5. Detail of features of the "Performance" tab.

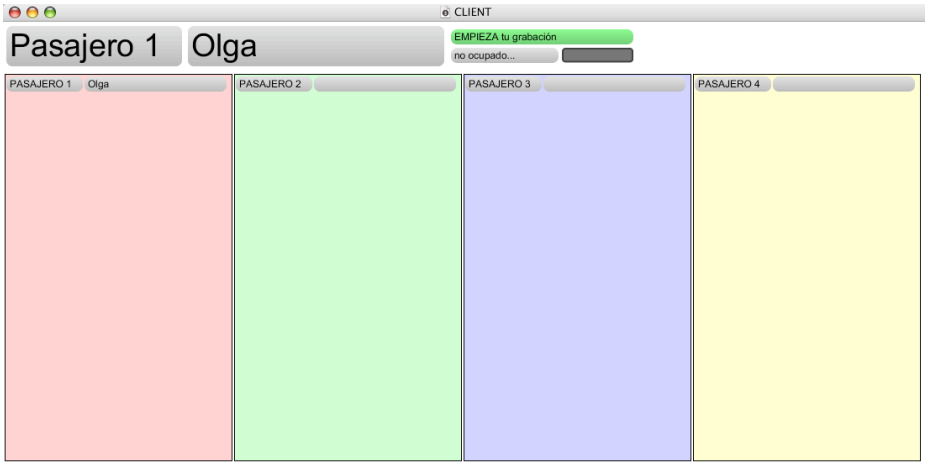


Figure 6. Client patch.

These aspects configure the network with an 'architecture' of a *synchronous* centralized network. Even if participants are stimulated by the sounds triggered by the leader, each one is free to record and playback their memories at any time, regardless the actions of the other participants and the leader. However they can influence rather than "modify [technically speaking] the music of their peers while it is being played"[18]. Spatialisation of soundscape and the use of participants' memories, are other important elements to add in its 'topology'.

Thus, in this hybrid sonic Interconnected Networked performance, interdependency, for musical and social achievements, relies not only on Weinberg's concepts, but also on the actions driven by listening and remembering, and by the connections established between the participants about a common sound structure: the soundscape. Sonic engagement is intrinsic to the process of remembering, in this context. It is strongly based on the ethnographic process, and on the emotional attitude of commuters toward this particular soundscape.

4. RESULTS - TYPES OF INTERACTION

In México DF, four groups of commuters improvised[8] in a large, non-isolated space. Their voices were amplified while recording. In this first improvisation, the amplitude of the playback of their voices differed from the amplitude of recording, which was a technical issue that needed improvement. In Paris, three groups of participants improvised[9] in a smaller space, with a more controlled acoustic. In order to improve the mentioned technical problem that occurred in México DF, participants in the performance in Paris - recorded without live amplification (i.e. 'secretly'), and played it back over the loudspeakers, thus discovering only later what the other participant had recorded. These two approaches defined an important difference between these two performances. Between México and Paris there were also certain changes in the server interface. Access to all the sounds recorded in the journey was possible in Paris, in contrast to the experience in México DF, where only the sounds previously chosen by the four participants were available. The change of this feature facilitated the leader's role in triggering the environmental sounds, enabling the mixing of all possible sounds provided by participants.

In both cities, the mixture of metro soundscapes and the intensity of multiple voices remembering created diverse narratives and stimulated the sharing of the commuting experience. These narratives varied according to cultural references and individual expressive and communicative interests. However, there were similarities in their narrative expectations and in the connection with the space. Participants switched between the "real space" (as if they were in the metro), their personal "memory space" (the memories that they share with their voices), and the "performance space", where the metro is perceived as a background space. These spaces are *Shared Sonic Environments* that bridge the experience of this acoustic community.

Performance forms that participants experienced can be divided and explained through the following typology:

5.1. Reflections about self and the space:

Participants made comments about the sounds that they were listening to, which also provoked responses from other participants. For example, in México, they expressed feelings related to the future of the city and the role of the metro: a shelter, a hub of popular culture, a chaotic city and its informal economy, and the role of the vendors. There were opinions about likes and dislikes of certain sounds and the role of the music in the journey. Some childhood memories were expressed, and also a memory of water. In Paris, comments included the weariness of the routine and the questioning and affirmations of Paris as a city. "Paris the beautiful city? Why Paris? It could be another city." (BN, my translation) "*This* is Paris and not the Eiffel Tower. This is Paris." (MM, my translation)

5.2. Sonic play:

Participants found musicality through abstraction, transforming everyday verbalization, through poetic sounds and words going beyond of the meaning of the words: "Avanzo, avanzo, avanzo, hacia abajo, hacia abajo, hacia abajo" (AI), "Des-cendre, des-cendre"(LW). Onomatopoeias were also common in the expression of the participants. They often imitated the sounds of machinery, which also lent rhythm and sonic textures to the experience, such as the case of airy sounds: "Shhhhhh, silence, silence, shhhhhhh"(MM), "Pfffff"(AS), or a screeching sound. In México, participants took advantage of one commuter's foreign language to imitate or play with the situation. The possibility of the immediate repetition of their own recordings tended to dominate the performance as a way of finding mixtures and rhythms. In some cases, this followed the rhythm of the metro, and in others it created a completely different soundscape, in a DJ fashion, which left the metro environment in the background. Within the journey, the sonic space of the carriage as it travels through a tunnel was a special place of engagement and intervention from participants playing with rhythms, and repetitions.

5.3. Incorporating everyday life:

Snippets of conversations between known or unknown people are common. Stories range between typical salutations: *"Doña Rosa dígame…"* ("Mrs Rosa tells me…", MR), "Hello Syl-vie" (GC) to non-sense interventions, which create interesting and humorous stories: *"esa mano es tuya?"* ("is this hand yours?", OC). In México a participant took advantage of a mobile

[8] At the Centro Multimedia, CENART, on the 26th of July, 2008. In the Sala Manuel Felguerez.

[9] At the Plate-forme Technologique, Maison des Sciences de l'Homme, MSH, Paris Nord, on the 24th of January, 2009.

phone call that he received during the performance, including it spontaneously in the improvisation. The other participants followed the conversation by making short comments that made the situation more humorous.

5.4. Story-telling / enacting:

Dramatic story-telling could take a conversational manner *"Oh la-la, suicide?"(GC), "No, ce n'est pas un suicide. C'est un chien, un chien sur la voie" (SD). (*"Oh dear, was it a suicide?" "No, it was not a suicide, it was a dog, a dog on the railway").

5.5. Imaginary spaces:

Participants created imaginary spaces into which they invited each other, for example, to dance "a waltz". Here time and space diverge from the routine, and this is prompted by the music that they are listening to. Music plays the role of linking with other spaces and times, and participants take advantage of this situation.

5.6. Communication Media:

Participants sent political messages, using the improvisation as a diffusion medium: *"Espero que la gente vaya a votar este domingo. No podemos dejar que PEMEX se quede en manos extrañas"* (EV). ("I hope people go to vote this Sunday. We cannot permit that PEMEX [Mexican Petroleum Company] is managed by foreign hands").

5.7. Just listening/intimate experience:

The feeling of not being obliged to record or play back was an unexpected option here. The setting, with a microphone and a big screen, can be intimidating for many people and particularly for non-performers. In the improvisation, some of them decided to simply listen. In Paris, one group was concerned with intimacy, and their memories were not played back often. A post-performance reflection by them was about respecting others' voices, or being self-conscious. This touched the boundaries of intimacy in the performance experience. They talked to themselves about dreams, the particular acoustic of the metro, and the anger felt about a robbery in the metro.

This typology reveals the performative strength of the improvisation, with its interweaving of narratives as a collective memory of an underground space. It is possible to see in these stories elements suggested by Williams, such as the aesthetics of underground environments moving between the sublime and the magical, derived from the realism given by the contemporary experience. Also, rhythmic relationships between machinery and the body are present in the performance, resembling experiences felt by London Underground commuters in previous research [1].

There are also perceived transformative possibilities for participants and their stories: transforming tragedy in daily life misunderstandings, exchange between unknown languages, and imagining a ballroom in a metro, amongst others. It's the interweaving of these stories, thanks to the networking structure and the switching between spaces, that makes it a special networked performance with extra-musical elements. An interconnection follows multiple purposes and causes, from the most individual to the most collective one. In the improvisation, we listen to "the unpredictable turns of chance permutation, the meatiness, the warmth, the simple, profound humanity of beings that bring presence and wonder to music."[9]

5. DISCUSSION WITH PARTICIPANTS

The variety of possible expressions combined with participants' diverse expectations, are considered part of the processes of remembering. The general feeling of the participants, who were listening and making sounds, was that it was an enjoyable yet strange experience. The experience of sharing in oneway or another was successful, and the making of linear or non-linear stories is part of the richness of this soundscape environment, transcending time and space. Conscious decisions of sounding and playing voices back create powerful moments of connection with the other participants in the exercise of playing together, creating tensions, harmonies and also disconnections.

Technical issues, such as the level of different voices, were discussed as potential obstacles in the performance. This issue defines whether participants are in the space of the metro or in a performance space. Some participants suggested it would be great if they were able to trigger their own (previously recorded and selected) sounds and not just their voices. Also, they would like to have a recording of the full improvisation for their records.

Acoustic conditions, such as the size of the room, were important and influenced the performances. The main obstacle described for participants in Paris was the fact they needed to write a "title" before recording. That interrupted the fluidity of remembering, recording, and playing back.

Expectations for people who are musicians[10] were different than for people who are not. Being able to control the sound amplitude and spatialisation were important issues for the former group. For other participants, narrative, in terms of story telling, imitation and repetitions, were some of the important, enjoyable issues.

Although it is important to highlight that participants in these two cities were already engaged with their own sounds and the process of remembering, some guests did join the improvisation. In México, this type of performance also took place in Morelia, a city four hours

[10] Some of the participants were musicians. In the call for volunteers the only requirement to participate was to be a commuter, regardless of their sonic or musical background.

away from the capital[11]. Volunteers improvisers who were not metro commuters (children, young people and adults with instruments) interacted in the setting, proving that the metro soundscape could engage people who are not familiar with this acoustic environment. Although improvising with instruments was not the purpose of this research, this kind of experience alludes to jazz forms. In the same vein, voices that play with repetitions and machinery sounds, suggest the influence of train and other mechanical and digital sounds as inspirations for the creation of musical and poetical forms, as has been the case in other contexts described by music scholars from soundscape and jazz history [7, 11, 13].

6. CONCLUSIONS AND FUTURE PLANS

The researcher believes that the improvisation "Listening and Remembering" is a catalyst for commuters from diverse backgrounds to perform an act of collective remembering about a shared soundscape: the underground public transport system. They use their voice as a source to *counter-point* the underground transport system's soundscape, in a broad sonic and musical experience. This opens scenarios to experiment with narratives and aesthetics related to the underground environment and to the rhythmical relationships between machinery and commuters' bodies. For example, the tunnel is certainly an important place (sonic and symbolic) that stimulates people's feelings and the sharing of experiences and sounds. It is also a space to link participants in the act of remembering. Here, poetic moments are powerful and take place within repetition, the rhythms of voices and machines, and the imitation of machines' sounds.

In light of Weinberg's proposed characteristics of Interconnected Musical Networks, this improvisation offers additional elements to the genre, such as spatialisation controlled by two sources (the leader, and the participants), and an emotional preparation of the commuters that is provided by the ethnographic process. These elements offer strength and singularity to it, and interdependency goes further than traditional musical expectations, expanding the notion of *Shared Sonic Environments* when participants locate themselves in either the "performance space", the "memory space" or the "real metro".

In terms of location, the future of the improvisation is two-fold: as an off-line performance, and as an on-line improvisation that integrates the elements developed in this experience. The off-line performance needs to improve conditions of spatialisation (soundscape and voices) to define the space where each participant would like to be. This can be done by including controls for amplitude, as previous improvisation experiences, such as Dilon's *jam2jam* project [5] with children and Weinberg's *Voice Networks* [19] with novices, have implemented. However, including more controls could also make this interface less simple for amateurs.

I do not intend to narrow the experience by establishing fixed characteristics of the space; it has been proved that the setting works for people with different expectations and skills. It also motivates the exploration of creativity to structure the performance in different ways. For this exploration, participants need time to get used to the setting, to be involved within the journey, and to relax, in order to start the creation of sonic narratives. This improvisation could also take the form of an installation environment off-line, where time and space are flexible for the participants during the sharing of their memories.

The option for participants to trigger their own metro sounds is important, as is the recording of the improvised sequence. The first, in order to decentralize the triggering of sounds of the soundscape; the second, as a publishing act and signature of authorship.

The online version of the improvisation requires technological development from scratch. Probably already existing packages and jamming environments[12] could be used to experiment with it. However the ideal is that the technology will be developed within the *Sounding Underground* environment.

While on-line environments can offer the feeling of being in one's own world, allowing uninhibited sounding expressions, the offline improvisation could also offer to participants a clearer statement about what to expect, particularly for the ones who are concerned with intimacy and the enjoyment of their own sonic space.

A combination of on-line and off-line improvisation might also be suitable for this project, having performance spaces in each city while interconnecting between cities, e.g. using an access grid. As noted previously, text entry is an obstacle in the performance space. However it is an important feature when working on-line, because it is integrated within the language of the Internet. Research in this area is needed to improve the performance space and to clarify for participants the role of the text. Future performances are planned using the sounds of the three metros involved (i.e. including London). It could ultimately be performed in any city.

The mixture of soundscapes, and feelings expressed in different languages and other voice expressions could integrate the three cities' soundscapes as a place symbolic of contemporary urban culture. And this could be offered to and performed by a variety of audiences, engaging them in the process of "Listening and Remembering". It is an improvisation that invites commuters to engage in a transformative process of sharing memories with their voices, widening for them the musical and poetic potential of this singular urban experience: the underground commuting routine.

[11] With the support of CMMAS (Mexican Centre for Music and Sonic Arts).

[12] Such as nin-jam http://ninjam.com/, or e-jamming http://ejamming.com/ .

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